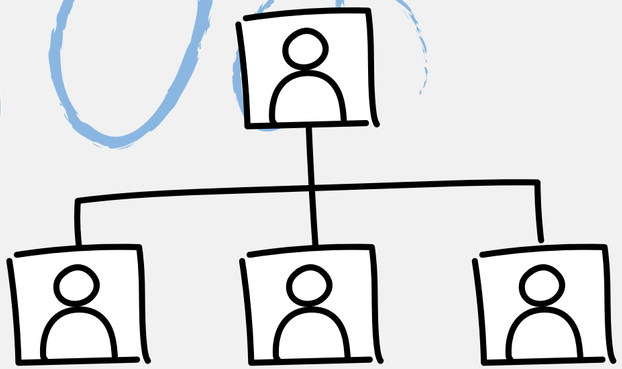
Sales Data Analysis Mini Project

Objective

- Understand the company's sales data
- Identify top-selling products
- Analyze sales performance by time periods and customers



team members

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Dataset Overview

- Number of rows and columns
- Key columns in the dataset:
  - ORDERNUMBER → Order ID
  - QUANTITYORDERED → Quantity ordered
  - PRICEEACH → Price per unit
  - SALES → Total sales
  - ORDERDATE → Order date
  - CUSTOMERNAME → Customer name
  - PRODUCTLINE → Product category
  - DEALSIZE → Deal size

Data Cleaning

01. Remove missing or incorrect values

02. Check and correct data types

03. Convert dates or numeric columns if needed

Data Analysis

Use tables or charts to show:

- Top-selling products
- Sales by month or quarter
- Top customers
- Sales by country or region

```
[13] import pandas as pd
     import numpy as np
     import matplotlib.pyplot as plt
```

```
[14] from google.colab import files
```

```
[15] uploaded = files.upload()
```

Choose Files mini_project.ipynb

- **mini_project.ipynb**(n/a) - 151535 bytes, last modified: 8/26/2025 - 100% done

```
Saving mini_project.ipynb to mini_project.ipynb
```

```
[16] df = pd.read_csv("/content/sales_data_sample.csv", encoding="latin1")
```

```
[17] print(df)
```

```
      ORDERNUMBER  QUANTITYORDERED  PRICEEACH  ORDERLINENUMBER    SALES  \
0           10107               30      95.70                2  2871.00
1           10121               34      81.35                5  2765.90
2           10134               41      94.74                2  3884.34
3           10145               45      83.26                6  3746.70
4           10159               49     100.00               14  5205.27
```

Variables Terminal

```
[21] df["ORDERDATE"] = pd.to_datetime(df["ORDERDATE"], errors="coerce")
```

```
[22] print(df["ORDERDATE"].head())
     print(df["ORDERDATE"].dtype)
```

```
0   2003-02-24
1   2003-05-07
2   2003-07-01
3   2003-08-25
4   2003-10-10
Name: ORDERDATE, dtype: datetime64[ns]
datetime64[ns]
```

```
[23] df_cleaned = df.dropna()
```

```
print(df_cleaned.head())
```

```
     ORDERNUMBER  QUANTITYORDERED  PRICEEACH  ORDERLINENUMBER    SALES  \
10         10223               37     100.00                1  3965.66
21         10361               20      72.55               13  1451.00
40         10270               21     100.00                9  4905.39
47         10347               30     100.00                1  3944.70
51         10391               24     100.00                4  2416.56
```

Variables Terminal

Commands + Code + Text Run all

```
[25] total_sales = df["SALES"].sum()
     print("Total Sales:", total_sales)
```

```
Total Sales: 10032628.85
```

```
avg_sales = df.groupby("ORDERNUMBER")["SALES"].sum().mean()
print("Average Sales per Order:", avg_sales)
```

```
Average Sales per Order: 32679.572801302937
```

```
[27] sales_mean = np.mean(df["SALES"])
```

```
[28] print("Mean Sales: ", sales_mean)
```

```
Mean Sales:  3553.889071909316
```

```
[29] sales_median = np.median(df["SALES"])
```

```
[30] print("Median Sales: ", sales_median)
```

```
Median Sales:  3184.8
```

```
[31] sales_std = np.std(df["SALES"])
```

```
[32] print("Standard Deviation Sales: ", sales_std)
```

```
Standard Deviation Sales:  1841.538852065539
```

```
[33] max_sales = df['SALES'].max()
```

```
[34] print("Max Sales: ", max_sales)
```

```
Max Sales:  14082.8
```

```
[35] min_sales = df['SALES'].min()
```

```
[36] print("Min Sales: ", min_sales)
```

```
Min Sales:  482.13
```

```
[37] sales_by_productline = df.groupby('PRODUCTLINE')['SALES'].sum()
```

```
[38] print("Sales for Amount of Qunatity Ordered: ", sales_by_productline)
```

Commands + Code + Text Run all

```
Trains                226243.47
Trucks and Buses     1127789.84
Vintage Cars         1903150.84
Name: SALES, dtype: float64
```

```
[39] sales_by_year = df.groupby("YEAR_ID")['SALES'].sum()
```

```
[40] print("Sales per Year: ", sales_by_year)
```

```
Sales per Year:  YEAR_ID
2003    3516979.54
2004    4724162.60
2005    1791486.71
Name: SALES, dtype: float64
```

```
[41] sales_by_dealsize = df.groupby("DEALSIZE")['SALES'].sum()
```

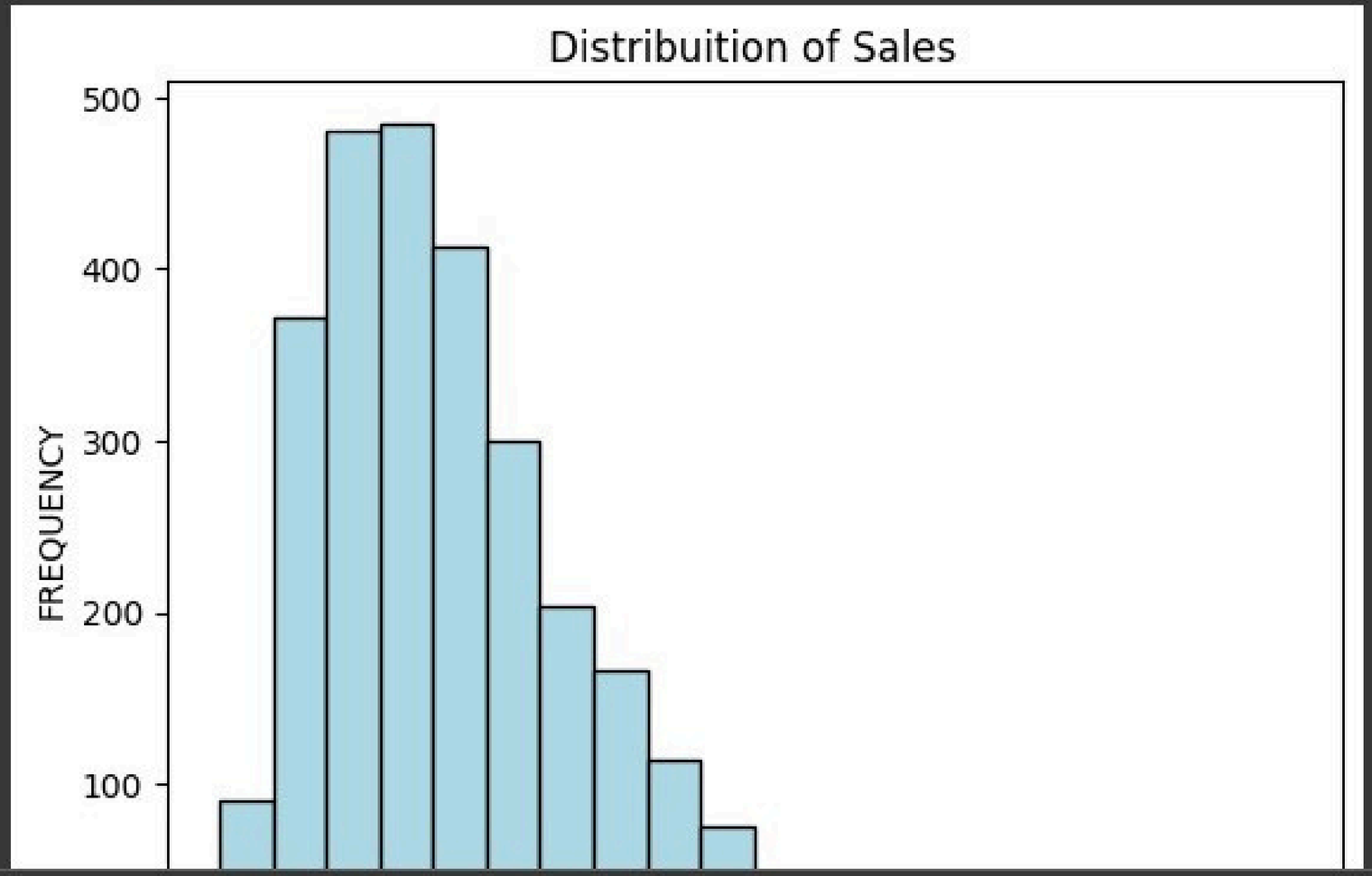
```
[42] print("Sales by Deal Size: ", sales_by_dealsize)
```

```
Sales by Deal Size:  DEALSIZE
Large     1302119.26
Medium    6087432.24
Small     2643077.35
Name: SALES, dtype: float64
```

Variables Terminal

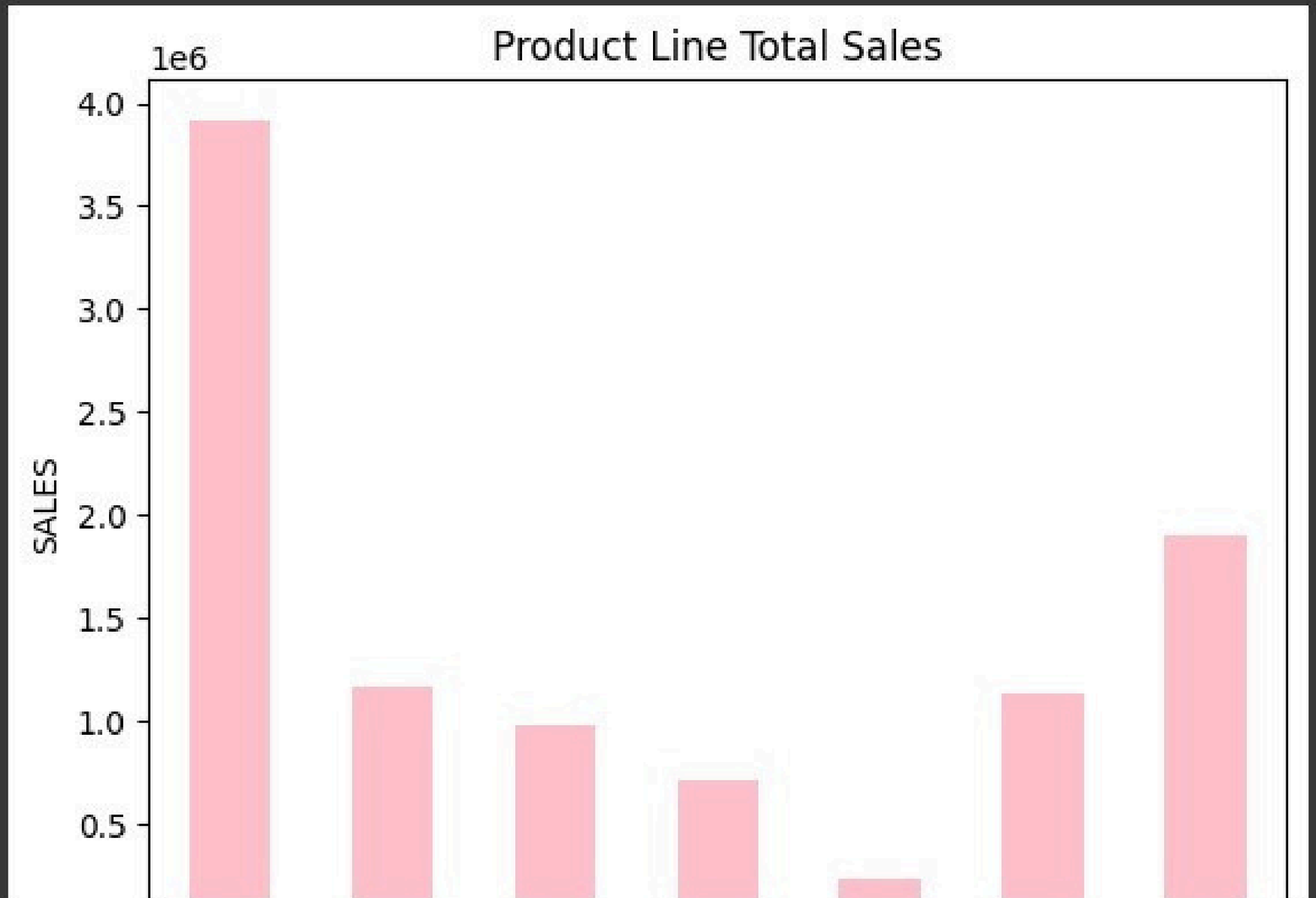
Visualizations

```
[43] plt.hist(df['SALES'], bins=20, edgecolor='black', color='lightblue')
     plt.title("Distribuition of Sales")
     plt.xlabel("SALES")
     plt.ylabel("FREQUENCY")
     plt.show()
```



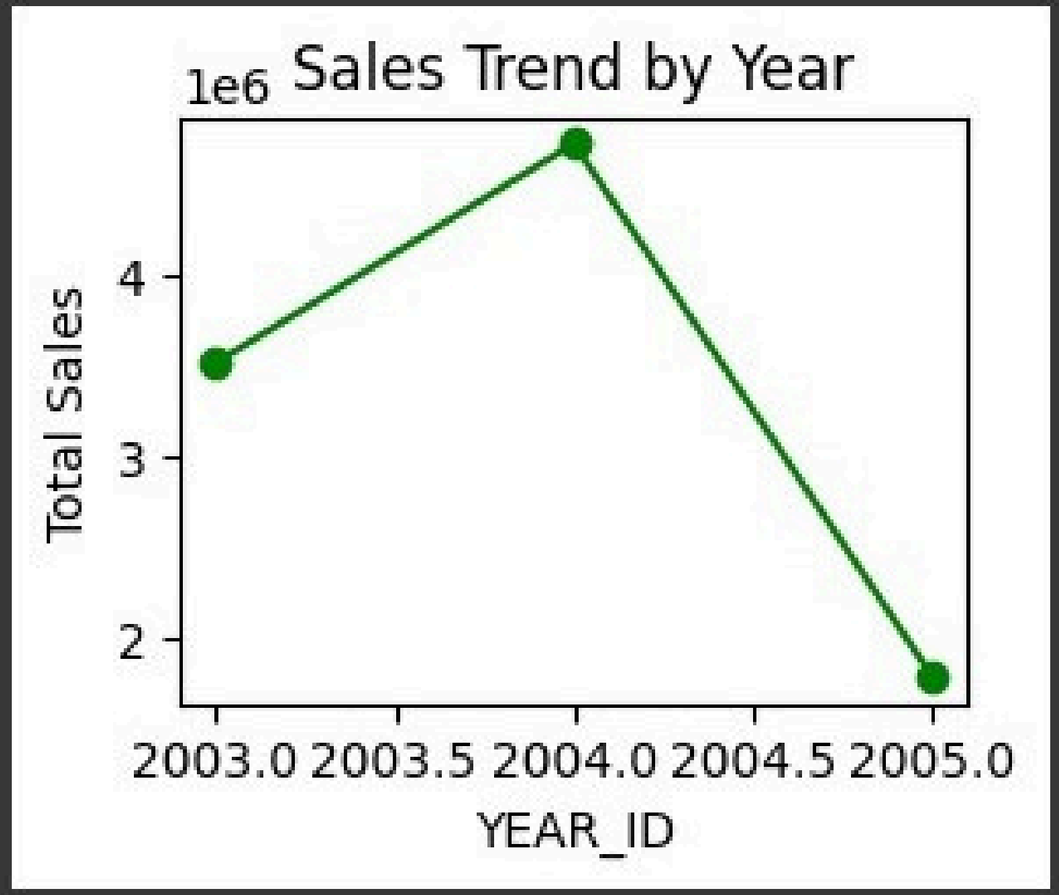
Visualizations

```
[44] sales_by_productline.plot(kind='bar', color='pink')
     plt.title("Product Line Total Sales")
     plt.ylabel("SALES")
     plt.show()
```

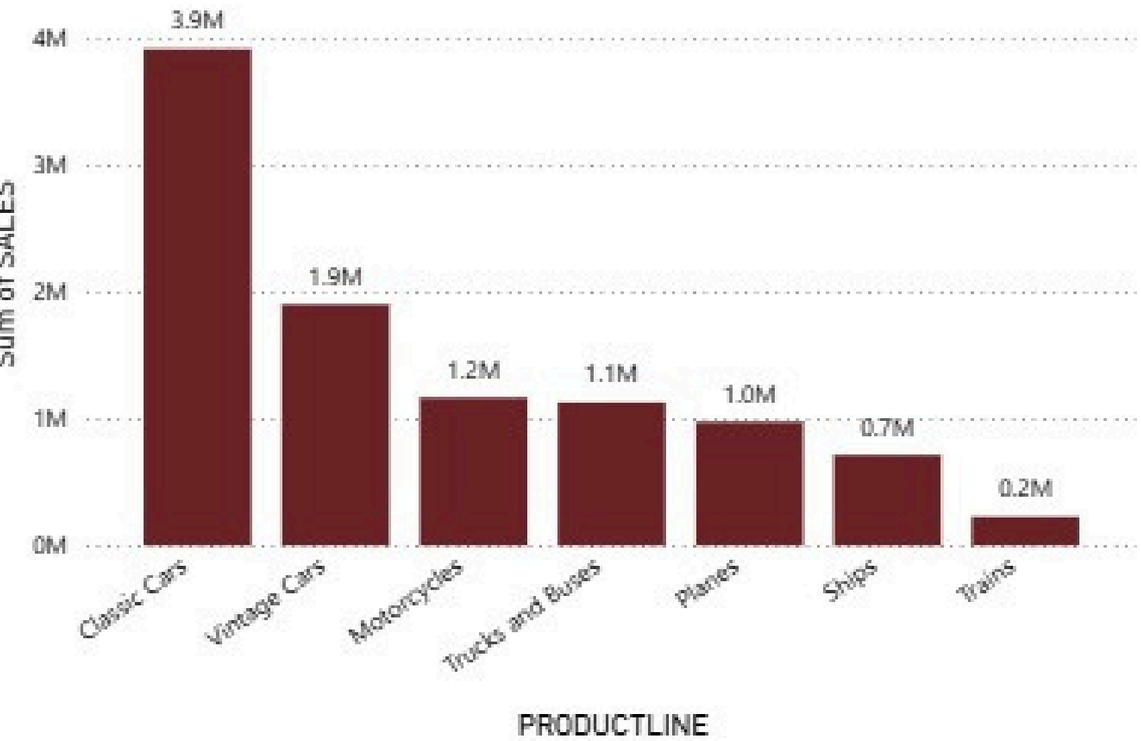


Visualizations

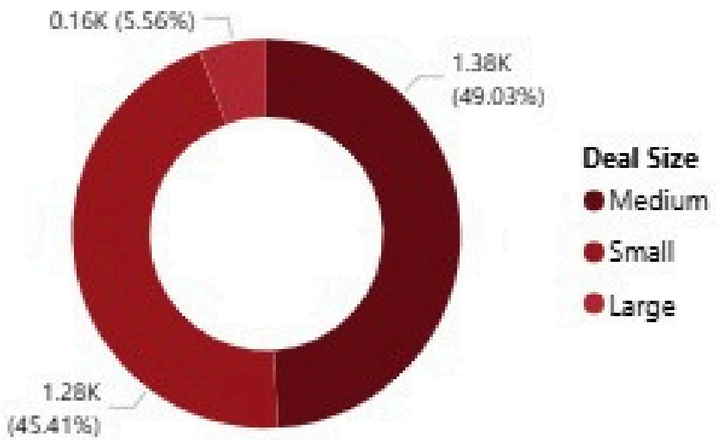
```
[45] plt.subplot(2,2,3)
     sales_by_year.plot(kind='line', marker='o', color='green')
     plt.title("Sales Trend by Year")
     plt.ylabel("Total Sales")
     plt.show()
```



Sum of SALES by PRODUCTLINE



Count of CUSTOMERS by DEAL SIZE

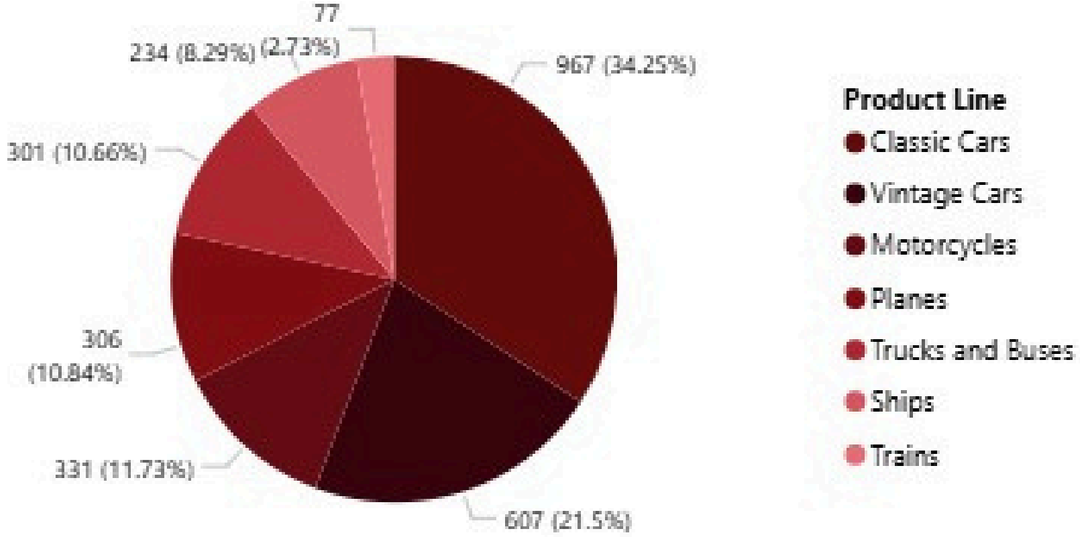


10.03M
Sum of SALES

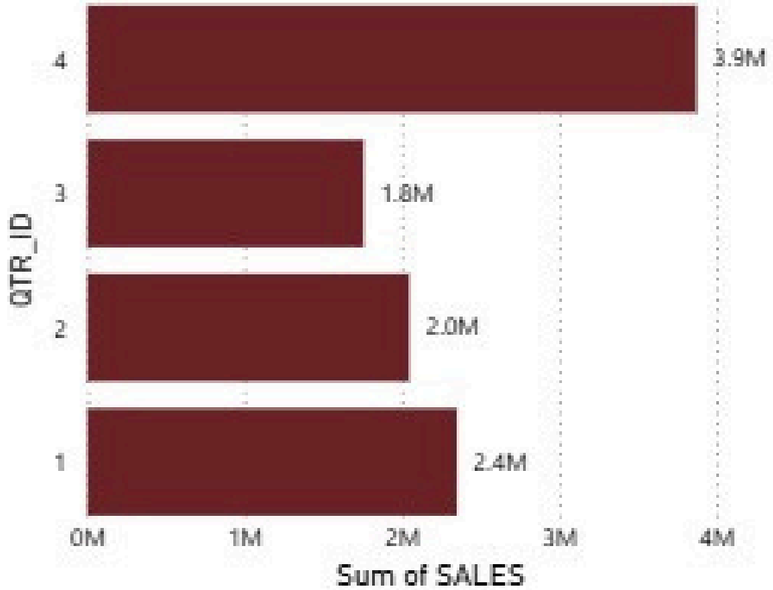
7
Count of PRODUCTLINE

2823
Count of CUSTOMERNAME

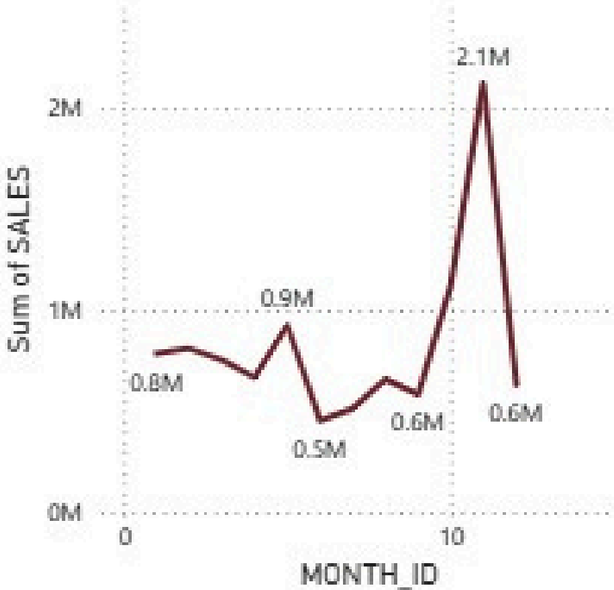
Count of CUSTOMERS by PRODUCT LINE



Sum of SALES by QTR ID



Sum of SALES by Month ID



Observation

- **Distribution of Sales (Histogram)**
- Most sales are concentrated between **1,000 and 5,000**.
- The peak (mode) occurs around **2,500**, meaning this is the most common sales level.
- The distribution is **right-skewed**, with a long tail towards higher sales.
- High-value sales above **7,000** are very rare.
- The **average sales value is likely higher than the median** due to a few extreme high values (outliers).

Insight: Sales are generally **low to medium**, with only a few large transactions driving up the average.

- **Product Line Total Sales (Bar Chart)**
- **Classic Cars** dominate total sales, contributing the highest revenue (>370,000).
- **Vintage Cars** come second (≈200,000), showing cars as the backbone of total sales.
- **Motorcycles** and **Trucks & Buses** perform moderately well (≈120,000 each).
- **Planes** have relatively lower sales (≈60,000).
- **Ships** and **Trains** show the weakest performance (<20,000).

Insight: The business heavily relies on **Classic and Vintage Cars**. Other product lines, especially **Ships and Trains**, may need strategic review, while **Motorcycles and Planes** could be potential growth opportunities.

……

- **Sales Trend by Year**
- Sales increased from **2003 to 2004** (~370K → ~420K).
- A **sharp decline** occurred in **2005** (~110K), indicating potential market or operational challenges.

- **Sales Distribution by Deal Size**
- The majority of sales come from **Medium deals** (61.6%).
- **Small deals** account for a significant portion (24.8%).
- **Large deals** contribute the least (13.6%), suggesting limited penetration in high-value clients.

Key Findings

Most sold products

Classic Cars

Vintage Cars

Periods with highest sales

2003 to 2004

Main customers

Medium-value customers

Sales are generally low to medium

business cases

1-Investigate and analyze the reasons behind the decline in sales in 2005

2-Develop a strategic sales plan focused on high-value and key customers.

3-Identify growth opportunities in the planes and motorcycles segments

4-Reinforce the automotive sector to sustain core revenue streams

Thank you very much!